Summary in Graph

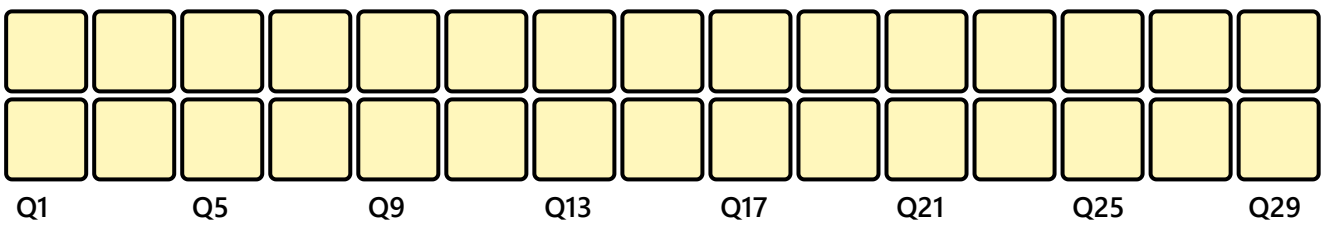
Exam Summary (GO Classes CS Test Series 2026 | Engineering Mathematics | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



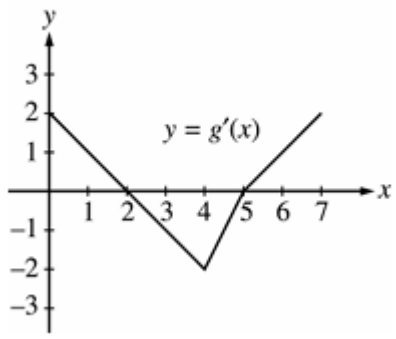
* Average Stats based on first-attempt stats of Top 10% (26) out of 262 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

The graph of the derivative g' of a function g with domain $[0, 7]$ is shown above. Which of the following must be true about g ?



I. g has a local maximum value at $x = 0$ and a local minimum value at $x = 4$.
II. g has a local maximum value at $x = 2$ and a local minimum value at $x = 5$.
III. $g(2) = g(5)$

A. I only
B. II only
C. III only
D. I and II

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Let

$$\lim_{x\to 0} \frac{\cos(3x)-1}{x^2} =$$

A. $\frac{9}{2}$

B. $\frac{3}{2}$

C. $-\frac{2}{3}$

D. $-\frac{9}{2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

If $y = x^3 + 3x^2 + 5$, then $\frac{dy}{dx} =$

A. $3x^3 + 6x^2 + 5$

B. $3x^2 + 6x$

C. $\frac{x^4}{4} + x^3 + 5x$

D. $15x^5$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type Award: 1 Penalty: 0.33 Others

Find $\frac{d}{dx}\int_1^x \left(t-t^4\right)dt.$

A. $x-x^4+1$

B. x^4-x+1

C. x^4-x

D. $x-x^4-1$

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #5 Multiple Choice Type Award: 1 Penalty: 0.33 Others

$\lim_{x\to 0}\left(\frac{2\sin 3x}{3\sin 2x}\right)=$

A. -1

B. 0

C. 1

D. ∞

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #6 Multiple Choice Type Award: 1 Penalty: 0.33 Others

Let

If $f(x)=2\sin(\cos x)$, then $f'(x)=$

A. $-2\sin x\cdot\cos(\cos x)$

B. $-2\cos x\cdot\sin(\sin x)$

C. $2\sin x\cdot\cos x$

D. $-2\sin x\cdot\cos x$

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #7 Multiple Choice Type Award: 1 Penalty: 0.33 Probability

Let X be a random variable with Poisson distribution $p(k;\lambda)$ Let $Y=(X+2)(X+1)$. What is the value of $E(Y)$?

A. $\lambda^2+3\lambda+1$
B. $\lambda^2+3\lambda+2$
C. $\lambda^2+4\lambda+2$

D. $3\lambda^2 + 3\lambda + 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

A rectangle has **length** X and **height** Y, where X and Y are **independent random variables**, each **uniformly distributed** over the interval $[0, 1]$.
Define:

- $A = XY$: the **area** of the rectangle
- $L = 2(X + Y)$: the **perimeter** of the rectangle

Which of the following statements are correct?

A. $\mathbb{E}[A] = \frac{1}{4}$

B. $\mathbb{E}[A] = \frac{1}{2}$

C. $\mathbb{E}[L] = 2$

D. $\mathbb{E}[L] = 1$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The lifetime (in years) of a certain type of electrical switch follows an exponential distribution with a **mean of 2 years.**
What is the probability that the switch **fails during the first year**?

A. $1 - e^{-1/4}$
B. $1 - e^{-2}$
C. $1 - e^{-1/2}$
D. $1 - e^{-1}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If $A = \begin{bmatrix} 1 & 0 & a & 2 \\ -2 & -1 & b & 3 \\ 3 & 1 & c & 1 \end{bmatrix}$ and the reduced row-echelon form of A is $R = \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, then $(a, b, c) =$

A. $(3, -2, 0)$

B. $(3, -4, 7)$

C. $(0, 0, 1)$

D. $(3, -2, 7)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 4×4 matrix such that $\det(A) = -2$. What is the value of $\det(\operatorname{adj}(A))$?

Your Answer: | Correct Answer: -8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If A and B are 3×3 matrices such that $\det(A) = 2$ and $\det(B) = -3$, then the value of $\det\left(A^2 B^{-1} A^T B^3\right)$ is

A. -72
B. 72
C. -108
D. 108

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 5×7 real matrix whose rows are linearly independent. What is the dimension of the null space of A ?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let B be a square matrix such that $C = \operatorname{adj}(B)$, where $\operatorname{adj}(B)$ denotes the adjugate of B. Suppose $|B| \neq 0$. What is the value of the determinant of the matrix $C^{-1}B^2$?

A. $|B|$

B. $|B|^2$

C. $\frac{1}{|B|^{n-3}}$

D. $\frac{1}{|B|^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a 4×4 matrix. If $\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$ are solutions of the system of linear equations

$A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$, then which of the following is also a solution?

A. $\mathbf{p} - \mathbf{q}$

B. $\mathbf{q} - 2\mathbf{p}$

C. $2\mathbf{p} - \mathbf{q}$

D. All of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

If a and b are positive numbers, then $\lim_{x \to 0} \left(1 + \frac{x}{a}\right)^{b/x} =$

A. e^{ab}

B. $e^{b/a}$

C. $e^{a/b}$

D. 1

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

Find the derivative of $y = \frac{\ln(2x^3)}{e^{2x}}$.

A. $\frac{3+2\ln(2x^3)}{e^{2x}}$

B. $\frac{3-2\ln(2x^3)}{e^{2x}}$

C. $\frac{3}{xe^{2x}} - \frac{2\ln(2x^3)}{e^{2x}}$

D. $\frac{3}{xe^{2x}} + \frac{2\ln(2x^3)}{e^{2x}}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

If $f(x) = \frac{x^3+2x-1}{x^2+x}$, then $f'(x) =$

A. $\frac{(x^2+x)(3x^2+2)-(x^3+2x-1)(2x+1)}{(x^2+x)}$

B. $\frac{(x^2+x)(3x+2)-(x^3+2x-1)(2x+1)}{(x^2+x)^2}$

C. $\frac{(x^2+x)(3x^2+2)-(x^3+2x-1)(2x+1)}{(x^2+x)^2}$

D. $\frac{(x^2+x)(3x^2+2)-2x(x^3+2x-1)}{(x^2+x)^2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X_1, X_2, \ldots, X_6$ be independent random variables such that

$$P(X_i = -1) = P(X_i = 1) = \frac{1}{2}, i = 1, 2, 3, \ldots, 6$$

Then $P\left[\sum_{i=1}^{6} X_i = 4\right]$ is

A. $\frac{3}{32}$

B. $\frac{3}{4}$

C. $\frac{3}{64}$

D. $\frac{3}{16}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $X_1, X_2, \ldots, X_{n+1}$ be independent random variables, each distributed as $\text{Bernoulli}(p)$.
Define

$$Y = \sum_{i=1}^{n} X_i X_{i+1}$$

Each term $X_i X_{i+1}$ equals 1 if and only if both X_i and X_{i+1} are 1 .

Which of the following statements are **true**?

A. Each term $X_i X_{i+1}$ is a Bernoulli random variable with parameter p^2.

B. The terms $X_i X_{i+1}$ are dependent due to overlapping variables.

C. The expectation of Y is np^2.

D. Each term $X_i X_{i+1}$ is independent, since all X_i are independent.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let E and F be two events with $P(E) > 0, P(F \mid E) = 0.3$ and $P\left(E \cap F^c\right) = 0.2$. Then $P(E)$ equals

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{4}{7}$

D. $\frac{5}{7}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two coins with probability of heads u and v, respectively, are tossed independently. If P (both coins show up tails) $= P$ (both coins show up heads), then $u + v$ equals

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X_1, X_2, X_3 and X_4 be independent random variables. Then which of the following pairs of random variables are independent?

A. $(X_1 + X_2, X_2 + X_3)$

B. $(X_1, X_1 + X_3)$

C. $(X_1 + X_2, X_3)$

D. $(X_2, X_1 + X_2 + X_3)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose A and B are events with $P(A) = 0.5, P(B) = 0.4$ and $P\left(A \cap B^c\right) = 0.2$. Then $P\left(B^c \mid A \cup B\right)$ is equal to

A. $1/2$

B. $1/3$

C. $1/4$

D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let E and F be two events with $0 < P(E) < 1, 0 < P(F) < 1$ and $P(E \mid F) > P(E)$. Which of the following statements is (are) TRUE?

A. $P(F \mid E) > P(F)$

B. $P\left(E \mid F^c\right) > P(E)$

C. $P\left(F \mid E^C\right) < P(F)$

D. E and F are independent

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A miner is trapped in a mine with **three doors**:

- **Door 1** leads to **safety** in **2 hours**.
- **Door 2** leads back to the mine after **3 hours**.
- **Door 3** leads back to the mine after **5 hours**.

Every time the miner faces the doors, he picks one **uniformly at random** (i.e., with equal probability). What is the expected time (in hours) for the miner to r**each safety**?

A. 6 hours
B. 9 hours
C. 10 hours
D. 12 hours

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The jobs arrive at a facility randomly over time. The number of jobs arriving in a fixed time interval follows which distribution?

A. Normal
B. Poisson
C. Exponential
D. Uniform

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE?

A. X and Y are not independent

B. Y and Z are dependent

C. Y and Z are independent

D. X and Z are independent

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $\{1, 2, 3, 4\}$ represent the outcomes of a random experiment, and $P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = 1/4$. Suppose that $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{3, 4\}$, and $A_4 = \{1, 2, 3\}$. Then which of the following statements is true?

A. A_1 and A_2 are not independent.

B. A_3 and A_4 are independent.

C. A_1 and A_4 are not independent.

D. A_2 and A_4 are independent.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

A is an $m \times n$ matrix and $\mathbf{b}$ is an $m \times 1$ vector. The equation $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. Consider the following statements:

(i) $m \leq n$

(ii) $n \leq m$

(iii) the rank of $A = n$

(iv) the rank of $A < n$

(v) $\det A = 0$

Which must be true?

A. only (i) and (v)
B. only (iv)
C. only (v)
D. only (iii) and (v)

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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